

Methods

Second-Level GLM (MATLAB)

Group-level inference on first-level GLM beta coefficients was performed in MATLAB using `nirs.modules.MixedEffects` (`MixedEffects.m`). The second-level model was:

$$\text{beta} \sim -1 + \text{group:experiment} + (1 \mid \text{subject}) \quad [\text{robust} = 1]$$

This intercept-free formulation estimated one beta per group-by-session combination (`calibacao`, `ac_01` ... `ac_09`) with a random subject intercept and robust estimation (`robust = 1`) to reduce leverage of outlying session estimates. The resulting `GroupStats` object provided group-level beta estimates for each channel, chromophore, and session.

Cortical Activation LMM (Python)

To model the longitudinal trajectory of cortical activation, a single LMM was estimated in Python 3.12 (`statsmodels` 0.14.6) on HbO beta coefficients aggregated by participant, session, and ROI. The model formula was:

$$\text{beta} \sim \text{group} * \text{sessao_num} + \text{roi} + (1 \mid \text{subject})$$

where `group` (reference: `nao_acelerado`), `sessao_num` (integer, 0-9; session 0 = calibration, serving as the intercept anchor), and ROI (Frontal vs Temporal; reference: Frontal) were fixed effects, and a random intercept per subject was included. Models were estimated using maximum likelihood (ML) with the BFGS optimiser (N = 280 observations: 14 subjects x 10 sessions x 2 ROIs). The same model structure was applied independently to HbO (primary chromophore) and oxygen saturation (StO₂, secondary analysis). Mann-Whitney U tests compared mean beta (averaged across training sessions 1-9) between groups for each ROI.

Results

Cortical Activation (HbO) — LMM

The LMM for HbO cortical activation yielded no statistically significant fixed effects (Table 3). The intercept (Non-Accelerated, Frontal, session 0) was $\text{beta} = -0.318$, $\text{SE} = 0.517$, $z = -0.615$, $p = .539$, 95% CI [-1.331, 0.695]. The group main effect was $\text{beta} = -0.126$, $\text{SE} = 0.693$, $z = -0.181$, $p = .856$, 95% CI [-1.484, 1.233]. The ROI effect (Temporal vs Frontal) was $\text{beta} = 0.463$, $\text{SE} = 0.329$, $z = 1.408$, $p = .159$, 95% CI [-0.182, 1.108]. Session progression was $\text{beta} = 0.036$, $\text{SE} = 0.081$, $z = 0.443$, $p = .658$, 95% CI [-0.123, 0.195]. The training-induced group x session interaction was $\text{beta} = -0.118$, $\text{SE} = 0.115$, $z = -1.034$, $p = .301$, 95% CI [-0.343, 0.106], indicating no significant differential HbO activation trajectory between groups. Mann-Whitney tests on mean HbO beta

confirmed the absence of between-group differences: Frontal, $U = 13$, $p = .165$, $r = 0.469$; Temporal, $U = 16$, $p = .318$, $r = 0.347$.

Table 3. LMM Results — Cortical Activation (HbO, N = 280)

Parameter	Beta	SE	z	p	95% CI
Intercept (Non-Acc., Frontal, Session 0)	-0.318	0.517	-0.615	.539	[-1.331, 0.695]
Group (Accelerated)	-0.126	0.693	-0.181	.856	[-1.484, 1.233]
ROI (Temporal vs Frontal)	0.463	0.329	1.408	.159	[-0.182, 1.108]
Session Progression	0.036	0.081	0.443	.658	[-0.123, 0.195]
Group x Session	-0.118	0.115	-1.034	.301	[-0.343, 0.106]

ML estimation; BFGS optimiser; Reference: Non-Accelerated, Frontal, Session 0.

Cortical Activation (StO2) — Supplementary

The StO2 LMM similarly yielded no significant effects. The group x session interaction was $\beta = -0.017$, $SE = 0.064$, $z = -0.269$, $p = .788$, 95% CI [-0.143, 0.109]. All other fixed effects were non-significant (all $p > .34$). Mann-Whitney: Frontal $U = 18$, $p = .456$, $r = 0.265$; Temporal $U = 14$, $p = .209$, $r = 0.429$. These results replicate the HbO findings and confirm the absence of differential group-level activation trajectories at the ROI level.